

Thermodynamic conditions control the valences state of semimetals thus affecting the behavior of PGE in magmatic sulfide liquids

Qing-Lin Liang^{a,b}, Xie-Yan Song^{a,*}, Richard Wirth^c, Lie-Meng Chen^a,
Song-Yue Yu^a, Nadezhda A. Krivolutskaia^d, Zhi-Hui Dai^a

^a State Key Laboratory of Ore Deposit Geochemistry, Institute of Geochemistry, Chinese Academy of Sciences, Guiyang, PR China

^b State Key Laboratory for Mineral Deposits Research, Department of Earth Sciences, Nanjing University, Nanjing 210023, China

^c German Research Centre for Geosciences GFZ, 3.5 Interface Geochemistry, Telegrafenberg, Potsdam, Germany

^d V. I. Vernadsky Institute of Geochemistry and Analytical Chemistry, Russian Academy of Sciences, 19 ul. Kosygina, Moscow, 119991, Russia

Received 18 August 2021; accepted in revised form 14 January 2022; Available online 22 January 2022

Abstract

In magmatic Nickel-Copper-Platinum-group elements (Ni-Cu-PGE) sulfide deposits, PGE occur as solid solution in base metal sulfides (BMS) or occur as platinum-group minerals (PGM). In different deposits, such as the world-class deposits at Jinchuan and Noril'sk district which formed from sulfide liquids with different oxygen fugacity (fO_2), the proportion of PGE forming PGMs and entering BMS as solid solution are variable and the PGMs are different in mineral species and morphology. The early-cumulated massive ores from the two deposits were analyzed using Electron Probe Microanalysis (EPMA), Laser Ablation Inductively Coupled Plasma Mass Spectrometry (LA-ICP-MS), Scanning Electron Microscope (SEM) and Focused Ion Beam and Transmission Electron Microscope (FIB-TEM). Combined with whole-rock PGE and semimetals data, this contribution discusses the effect of thermodynamic conditions (mostly fO_2) and semimetals on the distribution of PGE and the mineral species and morphology of PGM in magmatic sulfide deposits. The Noril'sk massive ores commonly contain 5–12 vol% of magmatic magnetite whereas magnetite is absent in the Jinchuan massive ores, which suggests that the sulfide liquid at Noril'sk had a higher oxygen fugacity (fO_2) compared to the sulfide liquid at Jinchuan. In the Jinchuan massive ores, only ~ 39 % Ir and 37 % Rh are presented in solid solution within BMS. Meanwhile, many (Ir-Rh-Pt)AsS crystals with grain sizes of 200 nm to 2.5 μ m were recognized enclosed in BMS grains and cobaltite-gersdorffite solid solution (CGSS) or rarely located at BMS grain boundaries. Microstructures of the (Ir-Rh-Pt)AsS minerals suggests that they probably crystallized from the sulfide liquid together with MSS. In contrast, more than 63 % Ir and 89 % Rh are presented in solid solution within BMS and no (Ir-Rh-Pt)AsS crystal was found in the Noril'sk massive ores. On the other hand, arsenides, stannides, antimonides, tellurides and bismuthides of Pd and/or Pt (Pt-Pd-semimetal minerals) were found in the Jinchuan and Noril'sk massive ores. This study suggests that thermodynamic conditions (especially fO_2) control the valence states of semimetals (especially As) affecting the behavior of PGE in sulfide liquids. A relatively low fO_2 of sulfide liquids leads to a relatively high As^{n-}/As^{n+} ratio, which is atomical ratio of the anion to cation of As (n represents valence number), and results in relative more Ir-AsS, Rh-AsS and Pt-As pre-nucleation clusters. These pre-nucleation clusters were enriched in or near the migrating MSS grain boundaries and finally reaching oversaturation and nucleation of (Ir-Rh-Pt)AsS crystals occurred. In contrast, a relatively high oxygen fugacity (fO_2) of sulfide liquids results in a low As^{n-}/As^{n+} ratio and Ir-AsS, Rh-AsS and Pt-As pre-nucleation clusters forming a negligible amount of (Ir-Rh-Pt)AsS minerals and facilitating Ir and Rh to enter MSS crystal lattice. As for the deposits formed from sulfide

* Corresponding author at: Institute of Geochemistry, Chinese Academy of Sciences, PR China.
E-mail address: songxieyan@vip.gyig.ac.cn (X.-Y. Song).

liquids with a relatively low fO_2 , such as the deposits at Jinchuan, Yangliuping and Sudbury, the fractional crystallization of MSS together with (Ir-Rh-Pt)AsS minerals resulted in PGE differentiation.

© 2022 Elsevier Ltd. All rights reserved.

Keywords: Thermodynamic condition; Semimetals; Platinum-group elements; Geochemical behavior; Magmatic sulfide deposits

1. INTRODUCTION

The platinum-group elements (PGE) are informative tracers for understanding petrogenetic processes such as planetary core/mantle differentiation and generation and evolution of the mantle derived basalts (Palme 2008; Harvey and Day, 2016). Particularly PGE compositions can provide basic insights into segregation and differentiation of sulfide liquids during the formation of magmatic sulfide deposits (Naldrett et al., 1992; Stekhin, 1994; Ballhaus et al., 2006; Mungall and Brenan, 2014; Barnes and Ripley, 2016). Experimental studies have shown that Rh and the IPGE (Os, Ir and Ru) are compatible with early crystallized monosulfide solid solution (MSS) whereas Pt, Pd and semimetals (As, Sn, Sb, Te, Bi) are incompatible with MSS and intermediate solid solution (ISS) and tend to concentrate in late residual sulfide liquids forming PGM at grain boundaries of sulfides (Cabri and Laflamme, 1976; Fleet et al., 1993; Barnes et al., 1997; Mungall et al., 2005; Helmy et al., 2010; Tomkins, 2010; Dare et al., 2014; Liu and Brenan, 2015; Distler et al., 2016; Sinyakova et al., 2017). It is commonly accepted that the differentiation of Rh and the IPGE from Pt and Pd in magmatic sulfide deposits is controlled by fractional crystallization of MSS (Keays and Crockett, 1970; Naldrett et al., 1992, 1996; Barnes et al., 1997; Mungall et al., 2005; Song et al., 2009). However, the IPGE-sulfarsenides found in sulfide ores from the deposits at Sudbury, Jinchuan and Yangliuping have been proposed to crystallize from sulfide liquids containing only 0.1 to 1 ppm (10^{-6}) of PGE and semimetals (Cabri et al., 1984; Dare et al., 2010a; Prichard et al., 2013; Liang et al., 2019). It was suggested that these IPGE-sulfarsenides played an important role in decoupling the IPGE and Rh from Pt and Pd in the Yangliuping deposit (Liang et al., 2019). Meanwhile, the proportions of PGE (mainly Ir and Rh) forming PGM compared with those occurring in base metal sulfides (BMS) as solid solution are variable in magmatic sulfide deposits (Cabri et al., 1984, 2003; Cabri and Laflamme 1976; Czamanske et al., 1992; Chen et al., 2015; Barnes and Ripley, 2016; Liang et al., 2019; Mansur et al., 2019b). Knowledge about the mechanism controlling the distribution of PGE and the mineral species and morphology of PGM in magmatic sulfide deposits is therefore important to better understand the behavior of PGE in natural sulfide liquids and the differentiation of PGE in magmatic sulfide deposits.

PGM are mostly compounds of PGE and semimetals (Cabri 1981, 2002) and PGE have a strong affinity for semi-metal melts (Helmy et al., 2013a; Piña et al., 2015, 2020). Semimetals, especially As, have variable valence states under different thermodynamic conditions (Canali et al., 2017; Maciag and Brenan, 2020). Helmy and Bragagni

(2017) found that the partition coefficient of Pt between MSS and the sulfide liquid decreased with the decreasing of fS_2 in Fe-Cu-S experimental systems, which was attributed to the increasing of As^{n-}/As^{n+} ratio and formation of Pt-As pre-nucleation clusters. Helmy and Bragagni (2017) suggested that Ir and Rh preferentially formed PGE-AsS pre-nucleation clusters whereas Pd and Pt formed semimetals pre-nucleation clusters in sulfide liquids. The Ir-AsS, Rh-AsS and Pt-As pre-nucleation clusters in the sulfide liquid were proposed to play an important role in the crystallization of (Ir-Rh-Pt)AsS and $PtAs_2$ minerals in deposits at Sudbury and Yangliuping (Dare et al., 2010b; Liang et al., 2019). The Pt and As concentrations in sulfide liquids at sperrylite saturation increased with the decrease of Fe concentration, thus Bai et al. (2017) suggested that fO_2 affects the valence state of As and the solubility of sperrylite. In summary, previous experimental studies suggested that thermodynamic conditions (fO_2 and fS_2) and semimetals contents in sulfide liquids influence the behavior of PGE. However, how thermodynamic conditions and semimetals in natural sulfide liquids affect the distribution of PGE and the mineral species and morphology of PGM in magmatic sulfide deposits is not well addressed.

The magmatic sulfide deposits at Jinchuan and Noril'sk district are two of the most important Ni-Cu-PGE deposits in the world (e.g. Genkin et al., 1981; Dyuzhikov et al., 1992; Stekhin, 1994; Naldrett et al., 1996; Naldrett, 2004; Song et al., 2006, 2009, 2012; Chen et al., 2013; Duran et al., 2020). It has been documented that the fO_2 of sulfide liquids of the Jinchuan and Noril'sk district are $\sim$ QFM + 1 and $>$ QFM + 1.5, respectively (Li et al., 2009; Barnes et al., 2013). Whole-rock compositions of PGE and semimetals of the early-cumulated massive ores from Jinchuan and Noril'sk district were measured. Electron Probe Microanalysis (EPMA), Scanning Electron Microscopy (SEM), Laser Ablation Inductively Coupled Plasma Mass Spectrometry (LA-ICP-MS), Focused Ion Beam (FIB) and Transmission Electron Microscope (TEM) techniques were used to examine the differences in the distribution of PGE and the mineral species and morphology of PGM between the early-cumulated massive ores from the two deposits. Based on these data we discuss the effect of thermodynamic conditions and semimetals on the behavior of PGE during early-stage crystallization of natural sulfide liquids and its implications on PGE differentiation in magmatic sulfide deposits.

2. GEOLOGICAL BACKGROUND

2.1. The Jinchuan deposit

The world-class Jinchuan Ni-Cu-PGE deposit is hosted by an ultramafic intrusion in the Longshoushan terrane,

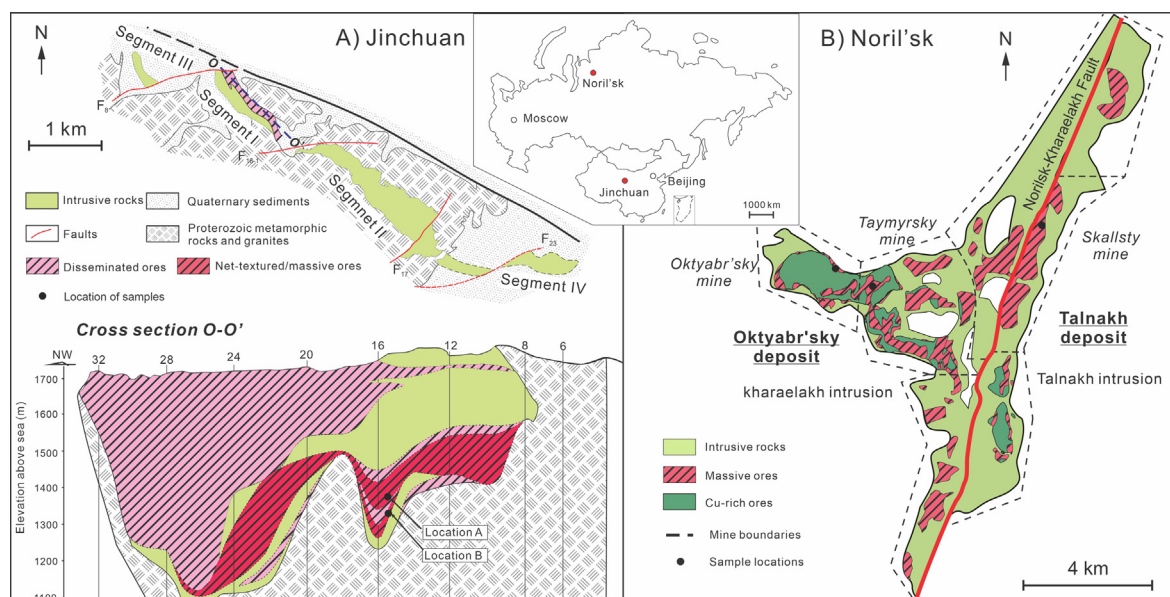


Fig. 1. **A.** Geologic map of the Jinchuan intrusion and cross section of orebody No. 24 with exploration lines annotated (modified from [Chen et al., 2013](#)). The massive ores selected for in-situ studies were collected from Location A (JC06-803) and location B (JC06-806, JC06-807 and JC06-809) ([Table S1](#)) by [Chen et al. \(2013\)](#). **B.** Plan view showing massive sulfides (including massive ores and Cu-rich ores) underlying the intrusions at Talnakh ore junction of the Noril'sk district (modified from [Stekhin, 1994; Naldrett, 2004](#)). The samples collected this time were from Oktyabr'sky mine, Skallsty mine and Taymyrsky mine. The massive ores selected for in-situ studies were from the Skallsty mine and Taymyrsky mine ([Table S1](#)). Orebody No. 24 of the Jinchuan deposit and Talnakh ore junction of the Noril'sk district are simplified as Jinchuan deposit and Noril'sk deposits below.

NW China ([Fig. 1A](#)) ([Song et al., 2012](#)). The three largest orebodies (No. 1, No. 2 and No. 24) contain > 90% of the total Ni, Cu and PGE reserves of the Jinchuan deposit ([Sixth Geological Unit, 1984](#)). Orebody No. 24 hosted in the lower unit of Segment I has higher PGE tenors than the No. 1 and No. 2 orebodies ([Song et al., 2009; Chen et al., 2013](#)). The thickness of orebody No. 24 varies from 20 to 130 m with net-textured ores overlain by disseminated ores ([Song et al., 2012; Chen et al., 2013](#)). Massive ores occur at the base of the net-textured ores ([Fig. 1A](#)) and small Cu-rich ore bodies occur within the footwall rocks ([Chen et al., 2013](#)).

2.2. The Noril'sk district

The Noril'sk district is the largest concentrations of chalcophile metals in the world and hosted by ultramafic–mafic sill-like intrusions at the northwestern margin of the Siberian Platform ([Dyuzhikov et al., 1992; Naldrett 2004; Lightfoot and Keays, 2005; Krivolutskaya et al., 2018](#)). The deposits are concentrated in several ore junctions. The Talnakh ore junction is the richest and comprises the Talnakh and Oktyabr'sky deposits hosted by the Talnakh and Kharalakh intrusions, respectively ([Fig. 1B](#)) ([Dyuzhikov et al., 1992; Stekhin, 1994](#)).

The disseminated ores of the Talnakh ore junction occur in the lower parts of the intrusions and the massive sulfides (with thickness up to 50 m) generally occur directly beneath the intrusions ([Dyuzhikov et al., 1992; Stekhin, 1994; Naldrett et al., 1996](#)). The massive sulfides of the Kharalakh intrusion show compositional zoning with massive ores at the edges and Cu-rich ores in the center ([Stekhin,](#)

[1994; Naldrett et al., 1996](#)). In the Talnakh intrusion, the compositions of massive sulfides vary from Cu–Ni massive ores in the north to Cu-rich ores in the south ([Fig. 1B](#)) ([Stekhin, 1994; Naldrett et al., 1996; Krivolutskaya and Sobolev, 2016](#)).

The geological background of the deposits and the mineralogical composition of sulfide ores at Jinchuan and Noril'sk district has been described in detail by [Genkin et al. \(1981\), Dyuzhikov et al. \(1992\), Stekhin \(1994\), Naldrett et al. \(1996\), Naldrett \(2004\), Song et al. \(2009, 2012\) and Chen et al. \(2013, 2015\)](#).

3. SAMPLES AND ANALYTICAL METHODS

Details of the samples ([Text S1](#)), methods for whole-rock compositions determinations and EPMA analyses ([Text S2](#)) are given in [supplementary materials](#).

The massive ores at Jinchuan and Noril'sk have Re–Os ages of 867 ± 75 Ma ([Yang et al., 2008](#)) and 247 ± 3.8 Ma ([Walker et al., 1994](#)), respectively, which are comparable with zircon SHRIMP U–Pb ages of 827 ± 8 Ma for the Jinchuan intrusion ([Li et al., 2004](#)) and 248 ± 4 Ma for the Noril'sk 1 intrusion ([Campbell et al., 1992](#)). Hydrothermal alteration commonly produces pyrite and anhedral magnetite along fractures of sulfide ores ([Barnes et al., 2006; Pina et al., 2013](#)), which were not observed in the Jinchuan and Noril'sk massive ores ([Fig. 2](#)). These indicates that the influence of fluid on the massive ores should have been minor.

No magmatic magnetite was found in the Jinchuan massive ores ([Fig. 2A–B](#)) whereas the Noril'sk massive ores commonly contain 5 to 20 vol% of euhedral to subhedral

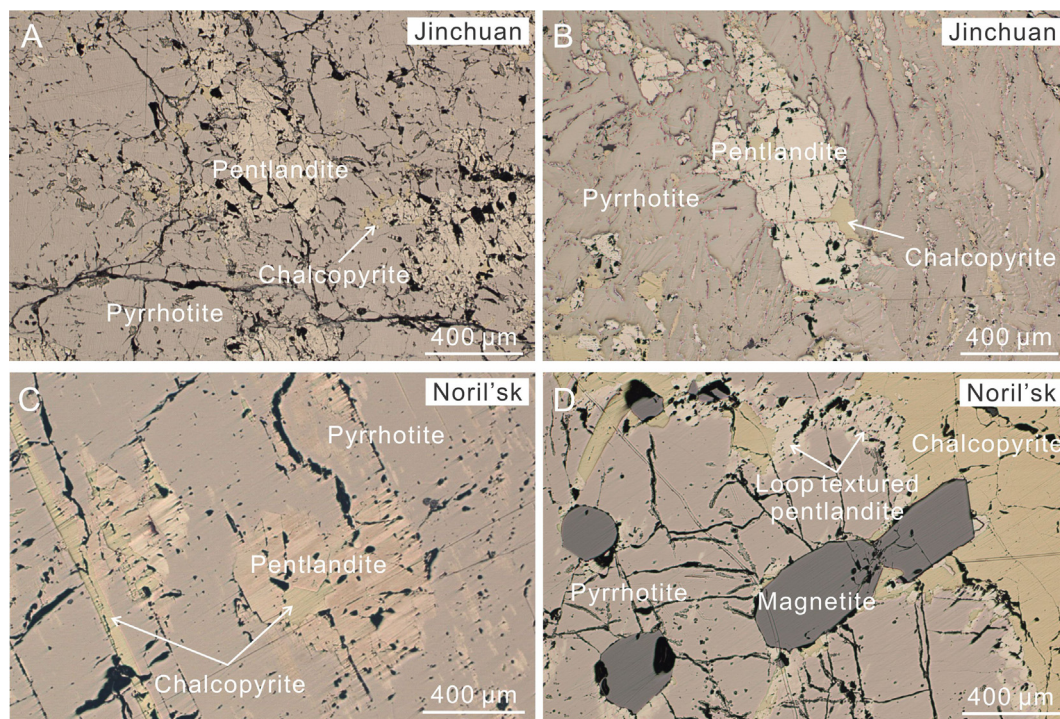


Fig. 2. The massive ores from the Jinchuan deposit (A–B) and the Noril'sk deposits (C–D). Magmatic magnetite is absent in the Jinchuan massive ores (A–B) while the euhedral to subhedral magnetites are common in the Noril'sk massive ores (D).

magmatic magnetites (Fig. 2D), which crystallized from the sulfide liquid (Genkin et al., 1981; Duran et al., 2020). Based on the fractional crystallization modeling (Fig. S1), four massive ores from No. 24 ore body of the Jinchuan deposit (simplified as Jinchuan deposit) and five massive ores from Talnakh ore junction of the Noril'sk district (simplified as Noril'sk deposits) with low Pd/Ru values representing the early-cumulated massive ores have been selected (Text S1) for *in situ* analyses. EPMA and LA-ICPMS measurements of a few BMS grains from the Jinchuan and Noril'sk massive ores did not show compositional variation from the cores to rims, indicating that the BMS grains selected for *in situ* analyses are fresh.

3.1. In-situ LA-ICP-MS analyses of BMS

Trace PGE and semimetals concentrations in the BMS of the Jinchuan and Noril'sk massive ores were measured using an Agilent 7700 × quadrupole ICP-MS coupled to an ASI RESOLUTION-LR-S155 laser microprobe at the Institute of Geochemistry, Chinese Academy of Sciences (IGCAS). The concentrations of PGE and semimetals in BMS are commonly only in ppb to ppm level. In order to improve the quality of LA-ICP-MS data, the PGE and semimetals were measured separately (separate locations). In a few spots they were analyzed in one location to capture the overlap between their peaks in time-resolved spectra of LA-ICP-MS (as shown in Fig. 3A). A laser beam spot size of 26 µm (diameter) was used for the measurements with a pulse rate of 5 Hz and a fluence of 3 J/cm². Ablations were performed in a He atmosphere at a 0.35 L/min flowrate.

The ablated aerosol was mixed with Ar (1.05 L/min) as transport gas.

⁵⁷Fe was used for internal standardization of BMS. The STDGL3, Po725, GSE-1 g and GED-1 g certified reference standards were used for external calibration of PGE and semimetals. MASS-1 was used to monitor the results. The isotopes measured and methods used for calibration and interferences corrections at the IGCAS were the same as those used at CODES, University of Tasmania, which were described in detail by Liang et al. (2019). Concentrations of PGE in the BMS of sample JC06-806 acquired at CODES (Chen et al., 2015) compare well with those at the IGCAS (Table S2; Fig. S3).

3.2. SEM-FIB-TEM analyses for PGM

The PGM in the massive ores were identified using a FEI Scios SEM at the IGCAS. The FIB-TEM measurements were done at the German Research Centre for Geosciences (GFZ), Potsdam. A Thermo Fisher Helios G4 UC-Dual Beam was used to prepare site-specific TEM foils following the method described by Wirth (2004). First, a ~ 2 µm-thick carbon layer was deposited at the selected area to avoid the contamination during Pt protective layer deposition. A FEI Tecnai G2 F20 transmission electron microscope (TEM) was used for chemical analyses and microstructural observations at the nanometer scale. The TEM is equipped with a field-emission gun electron source, a Gatan imaging filter GIFTM, an EDAX X-ray analyzer and a Fishione high-angle annular dark field (HAADF) detector. Nanoparticles of PGM with a diameter of ~ 1–2

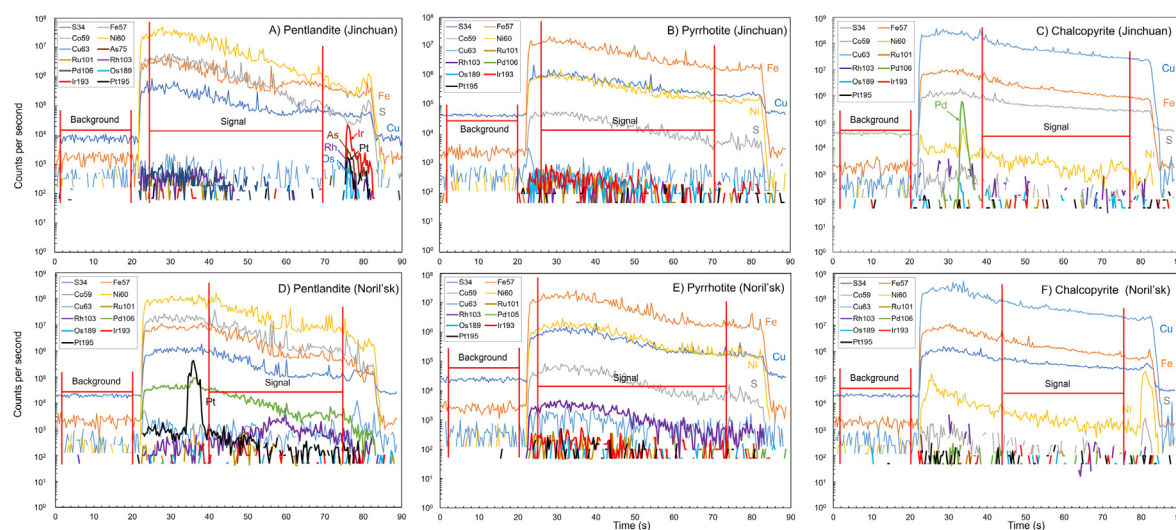


Fig. 3. Typical time-resolved spectra for Laser Ablation Inductively Coupled Plasma Mass Spectrometry (LA-ICP-MS) analyses of the BMS in the Jinchuan massive ores (A–C) and the Noril'sk massive ores (D–F). A. Time-resolved spectrum of pentlandite with an inclusion of Os-Ir-Ru-Rh-Pt-As. B. Time-resolved spectrum of pyrrhotite without inclusions. C. Time-resolved spectrum of chalcopyrite with a Pd mineral inclusion. D. Time-resolved spectrum of pentlandite with a Pt mineral inclusion. E and F. Time-resolved spectra of pyrrhotite and chalcopyrite without inclusions. Note that only in a few spots PGE and semimetals were analyzed together (Such as the spot shown in Fig. 3A).

nm could be easily observed in HAADF Z-contrast images. Bright-field and high-resolution TEM images were acquired as energy-filtered images applying a 20 eV window to the zero-loss peak of the electron energy loss spectra thus reducing image noise. HAADF images (Z-contrast) were acquired using a camera length of 75 mm to avoid collecting Bragg scattered electrons. Diffraction patterns were indexed using Inorganic Crystal Structure Database (ICSD).

4. RESULTS

4.1. Concentrations of PGE and semimetal elements in the BMS of the massive ores

In total 113 points were analyzed on BMS grains from the Jinchuan massive ores ($N = 4$) and 91 points from the Noril'sk massive ores ($N = 5$), respectively (Table S2). Most LA-ICP-MS time-resolved spectra show flat signals for PGE in BMS in solid solution (Fig. 3B, E, F) and a few time-resolved spectra show PGE-peaks when inclusions were encountered (Fig. 3A, C, D). In the Jinchuan massive ores, IPGE-Rh-Pt-(As) peaks were recognized in six time-resolved spectra from pyrrhotite and pentlandite (Fig. 3A) and only one Pd-peak was found in a time-resolved spectra of chalcopyrite (Fig. 3C). Similar peaks were reported in time-resolved spectra of the BMS from the Jinchuan massive and net-textured ores (Chen et al., 2015). However, in the Noril'sk massive ores, only one Pt-peak was found in time-resolved spectra of pentlandite (Fig. 3D) and no IPGE + Rh-peak was recognized in any time-resolved spectra of the BMS.

The integration intervals were selected to avoid peaks representing PGM and other inclusions. LA-ICP-MS results for the BMS from the Jinchuan and Noril'sk massive ores are listed in Table S2.

In the Jinchuan massive ores, pyrrhotite and pentlandite have similar concentrations of Os, Ru and Ir while the concentration of Rh is slightly higher in pyrrhotite than in pentlandite at a given concentration of Ir (Fig. 4A–C). However, pyrrhotite and pentlandite in the Noril'sk massive ores have comparable concentrations of Os, Ir, Ru and Rh (Fig. 4E–G). In both the Jinchuan and Noril'sk massive ores, Rh and the IPGE are enriched in pyrrhotite and pentlandite relative to chalcopyrite whereas Pd is concentrated in pentlandite and depleted in pyrrhotite and chalcopyrite (Fig. 4). The Pt and Pd concentrations of BMS from the Noril'sk massive ores are higher (Fig. 4H) compared to BMS from the Jinchuan massive ores (Fig. 4D).

Although Ru is positively correlated with Os in BMS from the Jinchuan massive ores (Fig. 4C), Ru and Rh only show approximate positive correlations with Ir (Fig. 4A–B). In contrast, Rh and the IPGE show positive correlations in the BMS from the Noril'sk massive ores (Fig. 4E–G).

The concentration of As in BMS from the Jinchuan and Noril'sk massive ores is commonly below the detection limit (Table S2) and occasionally up to 2.5 ppm As was found in Jinchuan pentlandite. In the Jinchuan massive ores, Te is below the detection limit in pyrrhotite and chalcopyrite but is enriched in pentlandite (~1–15 ppm). The concentration of Bi is similar in pyrrhotite and pentlandite (~0.01–2.0 ppm) and slightly higher in chalcopyrite (~0.4–3.7 ppm). In BMS from the Noril'sk massive ores, the concentrations of Te and Bi are mostly above detection limits and range from ~ 0.07 to 4.2 ppm and < 0.01 to 0.7 ppm, respectively (Table S2).

4.2. Mass balance calculation

The proportion of each PGE and semimetals hosted by each BMS was calculated following the method used by

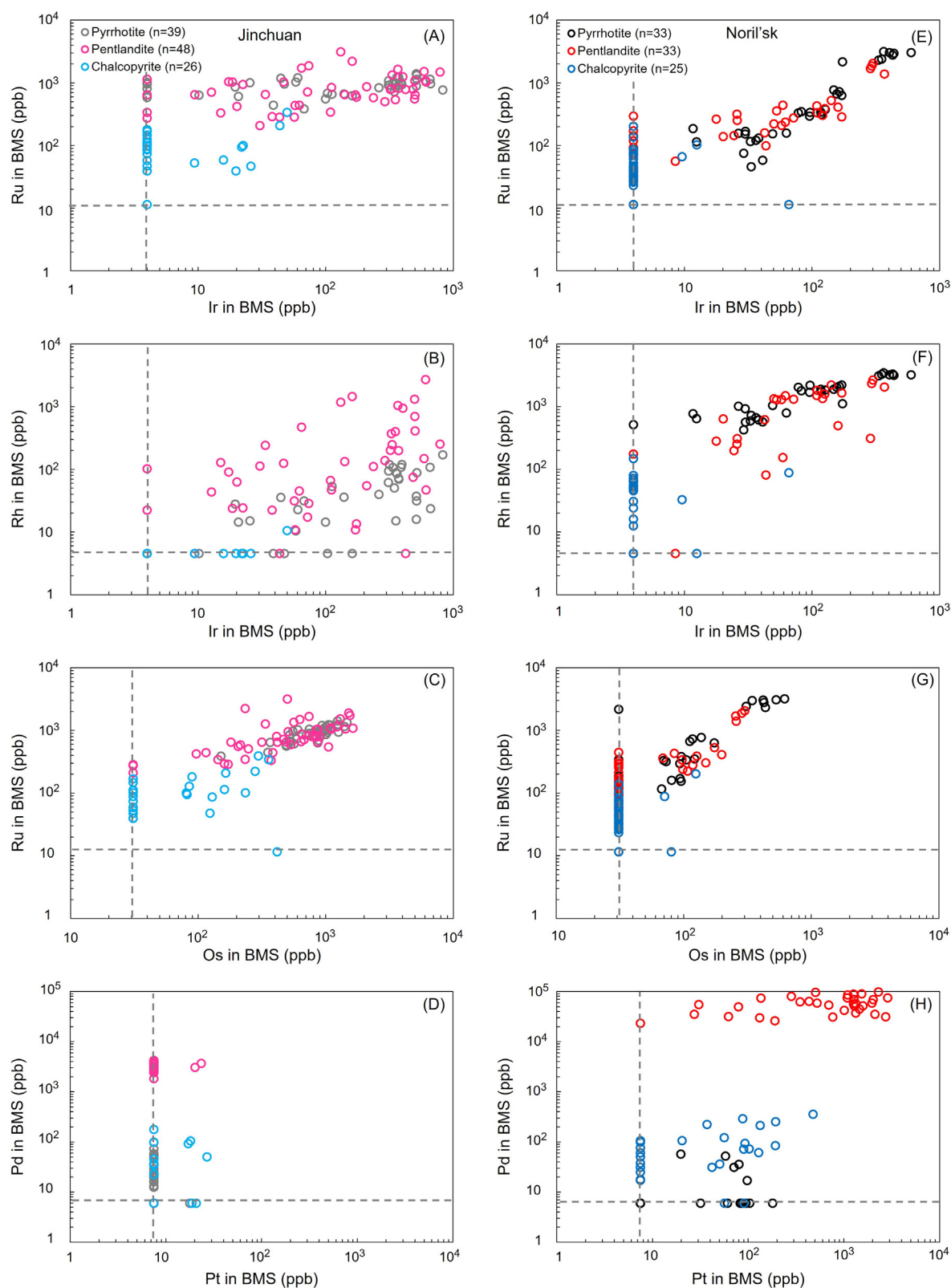


Fig. 4. Binary plots of Ir versus Os and Rh, Os versus Ru and Pt versus Pd in the BMS of the Jinchuan massive ores (A-D) and the Noril'sk massive ores (E-H). Dash grey lines indicate half detection limits for PGE by LA-ICP-MS. The LA-ICP-MS data are given in [Table S2](#). Abbreviations: n = the number of LA-ICP-MS analyses spots.

Barnes et al. (2008). The calculation requires the whole-rock concentration of each element (Table S1), the average concentration of each element in BMS and the calculated weight proportion of each BMS (Table S3). For the elements in BMS which are reported to be below the LA-ICP-MS detection limits (Table S2), half the values of the detection limits were used for the calculation of average concentration (Cabri et al., 2017). The weight proportion of pyrrhotite, pentlandite and chalcopyrite were calculated using whole-rock Cu, Ni and S (Table S1) and the composition of each BMS (Table S4), if all Cu formed chalcopyrite, Ni formed pentlandite and the remaining S formed pyrrhotite. The results of the mass balance calculation are shown in Fig. 5 and the primary data in Table S3.

The BMS from the Noril'sk massive ores are predominantly hosting Ir and Rh solid solution, accounting for 63% of the Ir and ~ 89% of the Rh, respectively (Fig. 5B). In contrast, only 39% of the Ir and 37% of the Rh occur in BMS as solid solutions in the Jinchuan massive ores (Fig. 5A). In both deposits, the BMS contain solid solutions of Ru (~100%), Pd (>81%) and Te (>55%) compared to small amounts of Pt (<15%), As (<5%), Sb (<17%) and Bi (<30%) (Fig. 5; Table S3).

4.3. Mineralogy of micrometer- and nanometer-sized platinum-group minerals

In the Jinchuan massive ores, 98 (Ir-Rh-Pt)AsS minerals, 1 erlichmanite ($\text{Os}(\text{Ru-Ir})\text{S}_2$), 32 CGSS and 9 Pt-Pd-semimetal minerals were found (Fig. 6A; Fig. 7A-C; Figs. S4A-I; Table S5A). In addition, 14 (Ir-Rh-Pt)AsS minerals and 26 Pt-Pd-semimetal minerals in BMS or at BMS grain boundaries of Jinchuan sulfide ores were found by Prichard et al. (2013). Euhedral to subhedral (Ir-Rh-Pt)AsS minerals (Fig. S5A) with grain sizes of 200 nm to 2.5 μm (up to ~ 6 μm) are heterogeneously distributed in pyrrhotite, pentlandite, CGSS and chalcopyrite or located at the BMS grain boundaries (Fig. 7A-C; Figs. S4A-G; Fig. S5B; Table S5A). Euhedral to subhedral CGSS crystals with grain sizes of 500 nm to 8 μm are enclosed in pyrrhotite, pentlandite and chalcopyrite or located at the grain boundaries of BMS (Fig. 7C; Figs. S4F-H; Table S5A).

The Pd-semimetal minerals ($n = 9$) are mostly subhedral to anhedral (Fig. S5C) with grain size up to 2.5 μm in pyrrhotite ($n = 7$), pentlandite ($n = 1$) and chalcopyrite ($n = 1$) (Figs. S4H-I; Table S5A).

In the Noril'sk massive ores, a total of 127 grains of cooperite (PtS), sperrylite (PtAs_2), Pt-Pd-semimetal minerals (mostly Pt-semimetal minerals), Ag-Au and Pt_3Fe alloy were found but no (Ir-Rh-Pt)AsS minerals were observed (Fig. 6B; Fig. 7D-F; Figs. S4J-R; Table S5B). This is supported by previous studies showing that PGE-sulfarsenides are rare in the Noril'sk deposits (Genkin et al., 1981; Czamanske et al., 1992; Kozirev et al., 2002; Spiridonov et al., 2015; Duran et al., 2017) and that Pt-semimetal minerals are the dominant PGM in massive ores (Duran et al., 2017). The PGM in Noril'sk occur mostly as single crystals (Fig. 7F; Figs. S4K, M and O-P) or as polymineralic assemblages (Fig. 7D-E; Figs. S4J, L, N and Q). PtS and Pt-Pd-semimetal minerals are commonly anhedral ($n = 65$) to subhedral ($n = 35$) and less commonly euhedral ($n = 17$) with grain sizes of 200 nm to 3 μm (up to ~ 10 μm) (Fig. 7D-F; Figs. S4J-Q; Fig. S5D). The subhedral to euhedral PtS and Pt-Pd-semimetal minerals ($n = 52$) are mostly enclosed in loop textured pentlandite (Text S1) and chalcopyrite or located at BMS grain boundaries ($n = 40$) (Fig. 7D-E; Figs. S4J, L-O and Q; Fig. S5E). Meanwhile, PtS and Pt-Pd-semimetal minerals in pyrrhotite ($n = 27$, Table S5B) are mostly anhedral ($n = 18$) (Figs. S4K and P; Fig. S5F).

4.4. Microstructures of platinum-group minerals

Seven TEM foils were cut across tiny (Ir-Rh-Pt)AsS minerals in BMS from the Jinchuan massive ores (Fig. 7A-B). From the Noril'sk massive ores, 3 TEM foils were cut from Pt-Pd-semimetal minerals in pyrrhotite and loop textured pentlandite (Fig. 7D-F) and 1 TEM foil was cut across the phase boundaries between pyrrhotite and pentlandite rich in PGE according to LA-ICP-MS analyses.

The (Ir-Rh-Pt)AsS minerals in the BMS from the Jinchuan massive ores are euhedral with grain sizes of 400 nm to 1.5 μm (Fig. 8A-C). These crystals have sharp phase boundaries and are heterogeneously distributed in pyrrhotite and

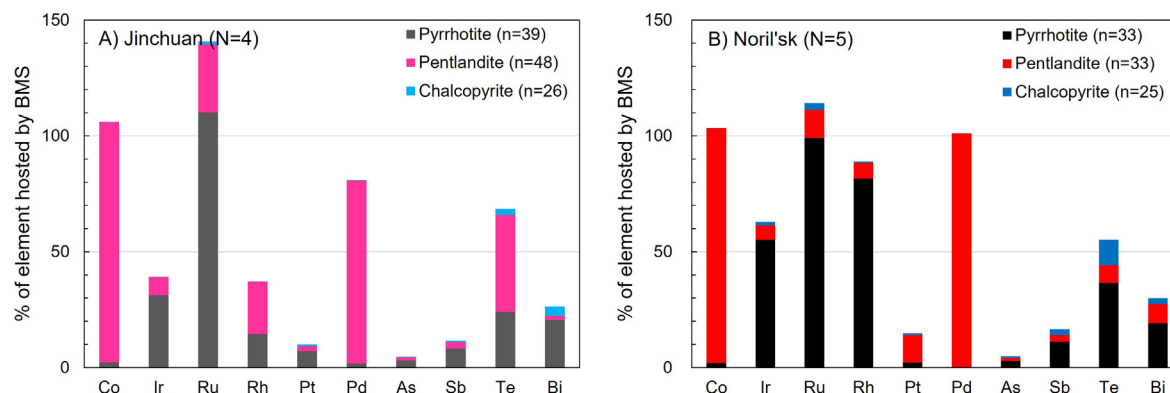


Fig. 5. Mass balance of the Co, PGE, As, Sb, Te and Bi in the BMS of the Jinchuan massive ores (A) and the Noril'sk massive ores (B). Each graph is plotted as the proportion of each element in pyrrhotite, pentlandite and chalcopyrite. The results of the mass balance calculation are shown in Table S3. Abbreviations: N = the number of the massive ores analyzed, n = the number of LA-ICP-MS analyses spots.

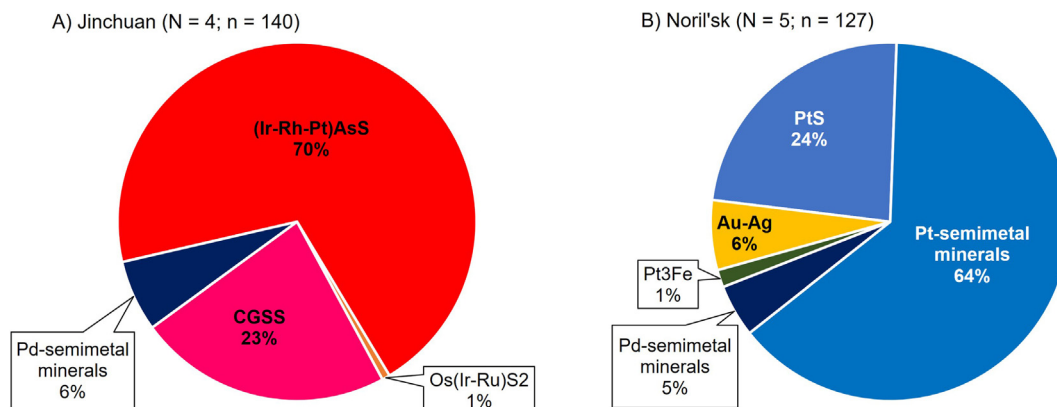


Fig. 6. Pie diagrams showing the relative proportions (%) of minerals (PGM, CGSS, Pt₃Fe and Au-Ag) identified by Scanning Electron Microscopy (SEM) in the Jinchuan massive ores (A) and the Noril'sk massive ores (B). The proportion of each kind of mineral are given in number and the source data are listed in Table S5. Abbreviations: N = the number of the massive ores analyzed, n = the number of the total PGM, CGSS, Pt₃Fe and Au-Ag grains, (Ir-Rh-Pt)AsS = irarsite-hollingworthite-platarsite solid solution, CGSS = cobaltite-gersdorffite solid solution, Os(Ir-Ru)S₂ = erlichmanite, PtAs₂ = sperrylite, PtS = cooperite, Au-Ag = electrum, Pt₃Fe = isoferroplatinum, Pd-Pt-semimetal minerals = Pt-Pd arsenides, stannides, antimonides, tellurides and bismuthides.

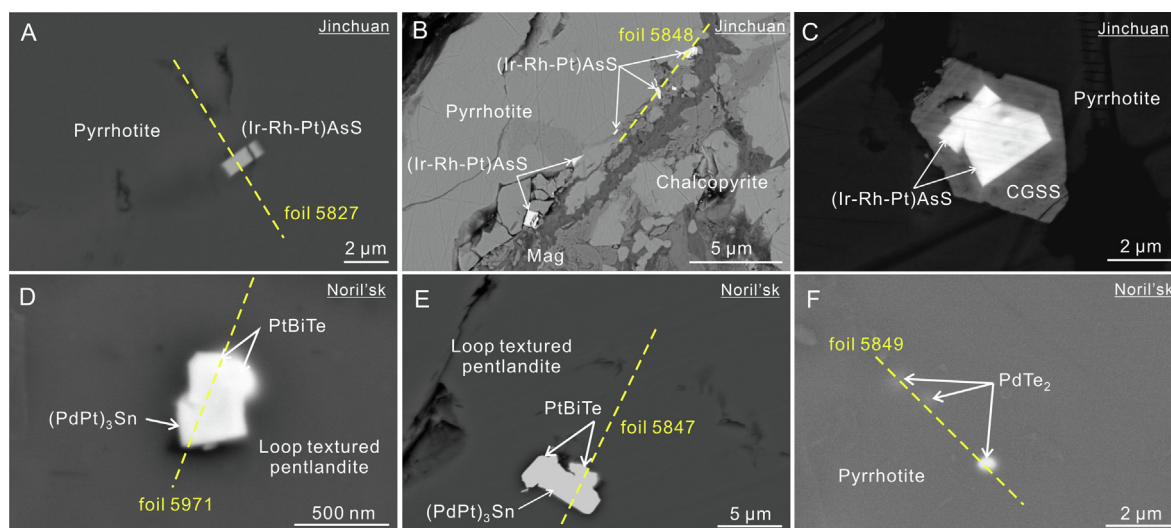


Fig. 7. Typical back scattered electron images of PGM in the Jinchuan massive ores (A-C) and the Noril'sk massive ores (D-F). **A.** Euhedral (Ir-Rh-Pt)AsS crystal in pyrrhotite. **B.** Cluster of euhedral (Ir-Rh-Pt)AsS crystals located at BMS grain boundaries. **C.** Two euhedral (Ir-Rh-Pt)AsS crystals in CGSS enclosed by pyrrhotite. **D.** Polyminerals associations of anhedral PtBiTe and euhedral (PtPd)₃Sn crystals in loop textured pentlandite. **E.** Polyminerals associations of PtBiTe and (PtPd)₃Sn crystals in loop textured pentlandite. **F.** Cluster of PdTe₂ crystals distributed in linear array in pyrrhotite. Yellow dash lines annotated with foil numbers indicate the positions for focus ion beam (FIB) sample preparations. Abbreviations: (Ir-Rh-Pt)AsS = irarsite-hollingworthite-platarsite solid solution, Mag = secondary magnetite, CGSS = cobaltite-gersdorffite solid solution, (PtPd)₃Sn = atokite, PdBiTe = michenerite, PdTe₂ = merenskyite, loop textured pentlandite = pentlandite formed through a peritectic reaction (Text S1).

chalcopyrite or located at the grain boundaries of BMS (Fig. 8A-C). There is no preferred orientation relationship between the (Ir-Rh-Pt)AsS minerals and the host BMS.

TEM studies revealed that polyminerals assemblages of Pt-Pd-semimetal minerals in BMS from the Noril'sk massive ores (Fig. 7D-E) consist of several individual anhedral to subhedral minerals with grain sizes varying from 300 nm to 1 μm (Fig. 8D-E). One of the PdTe₂ minerals with a diameter less than 100 nm enclosed in pyrrhotite (Fig. 7F) was identified as two phases consisting of an irregular-

shaped Pd₉Te₄ core and an unknown Pt-Pd-bearing telluride rim (Fig. 8F). Tens of nanoparticles around the PtBiTe mineral (Fig. 8D) and Pt-rich nanoparticles around the Pt-Pd-bearing telluride (Fig. 8F) were found in loop textured pentlandite and pyrrhotite matrices, respectively. No (Ir-Rh-Pt)AsS nanoparticles were found in pyrrhotite and pentlandite at the nanometer scale which contain up to 2723 ppb Rh and the IPGE. However, that could be a statistical problem because of the limited number of TEM foils studied.

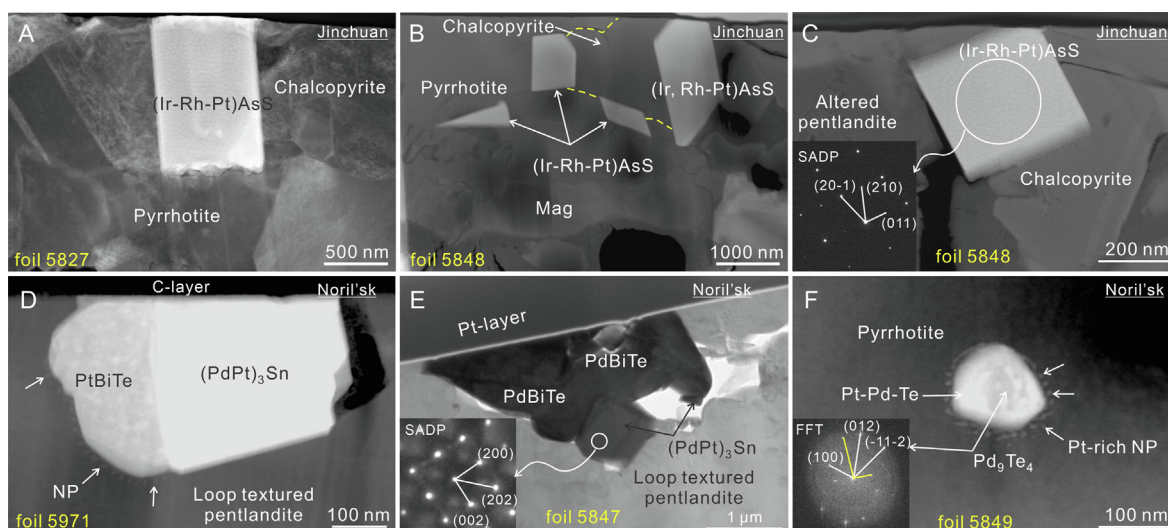


Fig. 8. Typical high-angle annular dark field (HAADF) images (A-D and F) and bright field image (E) of the PGM in foils from the Jinchuan massive ores (A-C) and the Noril'sk massive ores (D-F). Each image is annotated with foil number corresponding to the foil numbers in Fig. 7. A. Euhedral (Ir-Rh-Pt)AsS crystal in chalcopyrite. B. Euhedral (Ir-Rh-Pt)AsS crystals in pyrrhotite or at the phase boundaries of pyrrhotite and chalcopyrite. C. Euhedral (Ir-Rh-Pt)AsS crystal at the phase boundaries of chalcopyrite and altered pentlandite with selected-area diffraction pattern (SADP) of (Ir-Rh-Pt)AsS inserted. D. Polyminerall associations of subhedral PtBiTe and (PtPd)₃Sn crystals in pentlandite. E. Polyminerall associations of subhedral PdBiTe and (PtPd)₃Sn crystals in loop textured pentlandite with SADP of (PtPd)₃Sn inserted. F. One of the PdTe₂ mineral (Fig. 7F) was recognized as two phases in the TEM HAADF image. The inserted diffraction pattern was deduced from fast Fourier transforms (FFT) of high-resolution TEM image. The diffraction discs annotated by yellow lines indicate unindexed Pt-Pd-Te rim and the discs annotated by white lines was indexed as Pd₉Te₄ mineral. Note that there are tens of nanoparticles (NP) around the PtBiTe (D) and Pt-rich nanoparticles around the Pt-Pd-Te phase (F) indicated by white arrows. Abbreviations: (Ir-Rh-Pt)AsS = irarsite-hollingworthite-platarsite solid solution, Mag = secondary magnetite, CGSS = cobaltite-gersdorffite solid solution, (PtPd)₃Sn = atokite, PdBiTe = michenerite, PtBiTe = maslovite, Pd₉Te₄ = telluropalladinite, Pt-Pd-Te = unknown Pt and Pd-bearing telluride, loop textured pentlandite = pentlandite formed through a peritectic reaction (Text S1).

5. DISCUSSION

5.1. Constraints on the distribution of PGE in massive ores

The proportion of PGE entering BMS as solid solution as well as the species and morphology of the PGM are different in the Jinchuan and Noril'sk massive ores. About 39% Ir and 37% Rh occur in BMS as solid solution at Jinchuan (Fig. 5A) whereas 63% Ir and 89% Rh are present in solid solution within BMS at Noril'sk (Fig. 5B). Our results are supported by previous work that about 5% Ir and 46% Rh occur in BMS as solid solution at Jinchuan (Chen et al., 2015), whereas 92% Ir and 82% Rh are presented in solid solution within BMS at Noril'sk (Mansur et al., 2019b).

In the Jinchuan massive ores, the balance of Ir and Rh are found in PGM, which is consistent with IPGE-Rh-Pt-(As)-rich inclusions seen in the time-resolved spectra of LA-ICP-MS (Fig. 3A) and the (Ir-Rh-Pt)AsS minerals found using SEM (Fig. 6A; Fig. 7A-C; Figs. S4A-G; Table S5A). The (Ir-Rh-Pt)AsS minerals were identified using TEM to be irarsite-hollingworthite-platarsite solid solution (Fig. 8C). We proposed that the (Ir-Rh-Pt)AsS minerals account for the bulk of the Ir and Rh that were originally present in the sulfide liquid at Jinchuan, leaving only a small proportion in solid solution (Fig. 5A). Consequently, Ir shows poor correlations with Ru and Rh in BMS from the Jinchuan massive ores (Fig. 4A-B). We assume that in the

Noril'sk massive ores Rh and the IPGE are in solid solution in BMS based on the signals in LA-ICP-MS showing flat time-resolved spectra (Fig. 3D-F), and strong positive correlations for all Rh and the IPGE in the BMS (Fig. 4E-G). This conclusion is further supported by the TEM studies where no Rh and IPGE-bearing nanoparticles were found in pyrrhotite and pentlandite with up to 2723 ppb Rh and the IPGE, although the presence of atomic-scale clusters of IPGE and Rh cannot be ruled out.

The distribution of Pt, Pd and semimetals is similar in the Jinchuan and Noril'sk massive ores. The bulk of the Pt, Sb, Bi and minor amounts of the Pd and Te (Fig. 5) formed PtS and Pt-Pd-semimetal minerals (Fig. 7D-F; Figs. S4H-Q; Tables S5A-B).

5.2. Implications from the microstructure of PGM for petrogenesis

It has been suggested that PGM enclosed in BMS grains have formed by exsolution (Makovicky et al., 1986; Ballhaus and Ulmer, 1995; Makovicky, 2002; Barnes et al., 2008; Mansur and Barnes, 2020), they have crystallized directly from sulfide liquids (Hutchinson and McDonald, 2008; Dare et al., 2010b; Prichard et al., 2013; Liang et al., 2019) or they have crystallized from Pt, Pd and semimetal enriched residual liquid as the crystallization of MSS and ISS (Cabri and Laflamme, 1976; Dare et al., 2014; Godel et al., 2010; Tomkins, 2010).

With the crystallization of MSS, residual sulfide liquids were enriched in Cu, Pd, Pt and semimetals (Barnes et al., 1997; Mungall et al., 2005; Helmy et al., 2010; Liu and Brennan, 2015). The MSS cumulates at Jinchuan and Noril'sk have trapped a small amount of the residual sulfide liquid from which ISS has crystallized. During cooling, MSS and ISS predominantly exsolved to pyrrhotite, pentlandite and chalcopyrite (Cabri, 1973; Kullerud et al., 1969). Meanwhile, the loop textured pentlandite showing zonation in IPGE, Rh and Pd formed by peritectic reaction have been recognized in Noril'sk deposit (Mansur et al., 2019a). In the Jinchuan massive ores, the heterogeneously distributed nanometer- to micrometer-sized euhedral (Ir-Rh-Pt)AsS minerals are enclosed not only in pyrrhotite, pentlandite and CGSS, but also in chalcopyrite or are located at BMS grain boundaries (Fig. 7A-C; Fig. 8A-C; Figs. S4A-G; Table S5A). No preferred orientation relationships were found between the (Ir-Rh-Pt)AsS minerals and the host sulfides (Fig. 8A-C). Meanwhile, a common crystallographic orientation is also absent from these (Ir-Rh-Pt)AsS crystals (Fig. 8B). These observations suggest that the (Ir-Rh-Pt)AsS minerals probably have crystallized directly from the sulfide liquid, rather than having formed by exsolution, which resulted in oriented lamellae or sphere within host minerals (Putnis, 1992). Ru-Rh-Pt-arsenide nanoparticles having crystallized directly from sulfide liquids were also found enclosed in pyrrhotite from the Merensky Reef, Bushveld (Wirth et al., 2013).

Palladium, Pt and semimetals are incompatible with MSS. Therefore, they were enriched in or near the migrating MSS grain boundaries during grain growth (Barnes et al., 1997; Mungall et al., 2005; Helmy et al., 2010; Liu and Brennan, 2015; Distler et al., 2016; Sinyakova et al., 2017; Helmy and Botcharnikov, 2020). An enrichment of incompatible elements in or at migrating interfaces followed by nucleation and crystallization of the according thermodynamic stable phase was already described from other systems (silicate melt-apatite interfaces and fluid-magnetite interfaces) and seems to be a common mechanism (Gottesmann and Wirth, 1997; Yin et al., 2019, 2022). The subhedral to anhedral PtS and Pt-Pd-semimetal minerals enclosed in pyrrhotite at Jinchuan (Fig. S4I; Table S5A) and Noril'sk (Figs. S4K and P; Table S5B) have crystallized in or near the migrating grain boundaries and then have been overgrown by the MSS. The PtS and Pt-Pd-semimetal minerals in chalcopyrite and loop textured pentlandite at Noril'sk (Fig. 7D-E; Fig. 8D-E; Figs. S4J, L-O and Q) crystallized during the crystallization of ISS (Genkin et al., 1981; Cabri and Laflamme, 1976; Dare et al., 2014; Sinyakova et al., 2017) and that loop textured pentlandite formed through a peritectic reaction between MSS and evolved sulfide liquids (800 °C to 870 °C) (Mansur et al., 2019a; Barnes et al., 2020). These interpretations are supported by experimental studies showing that Pd and Pt formed stable phases with semimetals at relatively low temperatures (~500 °C to 900 °C and up to 1180 °C) compared to PGE-sulfarsenides (>1200 °C) (Cabri and Laflamme, 1976; Makovicky, 2002 and references therein; Helmy and Botcharnikov, 2020). Nevertheless, we can't exclude the possibility that Pt-rich

nanoparticles around Pt-Pd-bearing tellurides (Fig. 8F) have formed by sub-solidus exsolution (Makovicky et al., 1986; Barnes et al., 2008; Godel and Barnes, 2008; Mansur and Barnes, 2020).

5.3. Factors affecting the behavior of PGE in sulfide liquids

5.3.1. Differences in fO_2 and fS_2 between the sulfide liquids that have formed the Jinchuan and Noril'sk deposits

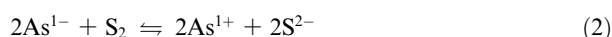
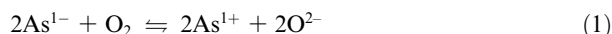
Duran et al. (2020) proposed that magmatic magnetite in sulfide ores of the Noril'sk deposits began to crystallize from a sulfide liquid at ~ 1050–1100 °C with fO_2 higher than QFM + 1.5 (Li et al., 2009). In contrast, the magmatic magnetite-free Jinchuan massive ores formed from a sulfide liquid with fO_2 close to QFM + 1 (Barnes et al., 2013). According to the method proposed by Toulmin and Barton (1964), calculations using the composition of pyrrhotite (Table S4) show that the fS_2 of the sulfide liquid at Jinchuan was relatively higher than that at Noril'sk in a temperature range from 600 to 1200 °C (Fig. S6), which is compatible with differences in fO_2 . Thus, the sulfide liquid at Jinchuan had a higher fS_2 (Fig. S6) but a lower fO_2 (~QFM + 1) (Barnes et al., 2013) compared to the sulfide liquid at Noril'sk ($fO_2 > \text{QFM} + 1.5$) (Li et al., 2009).

5.3.2. The effect of fO_2 and fS_2 and semimetals on the behavior of PGE

The solubility of PGE in MSS and the partition coefficients of PGE between MSS and sulfide liquid ($D_{PGE}^{MSS/Sul}$) increase with the increasing of fS_2 (Ballhaus and Ulmer, 1995; Barnes et al., 1997; Makovicky, 2002 and references therein). Whereas fO_2 has minimal influence on $D_{PGE}^{MSS/Sul}$ at fixed fS_2 (Mungall et al., 2005). Although the sulfide liquid at Noril'sk had a lower fS_2 (Fig. S6), the BMS contain a higher proportion of Ir and Rh as solid solution compared to the BMS from Jinchuan (Fig. 5). Thus, the distribution differences of Ir and Rh in the Jinchuan massive ores (Fig. 4A-C; Fig. 5A) and the Noril'sk massive ores (Fig. 4E-F; Fig. 5B) were not controlled directly by the fS_2 and fO_2 of sulfide liquids.

PGM are mostly compounds of PGE and semimetals (Cabri 1981, 2002). Semimetals can have different valence states depending on the thermodynamic conditions. For example, As may form stable bonds as As^{5+} , As^{3+} , As^{1-} and As^{3-} and the As^n/As^{n+} ratios vary depending on the fO_2 and fS_2 of sulfide liquids (Deditius et al., 2008; Liu and Brennan, 2015; Bai et al., 2017; Canali et al., 2017; Helmy and Bragagni, 2017; Maciag and Brennan, 2020). In experimental studies, PGM nanonuggets (such as $PtAs_2$ and $FePt$) had been commonly found in quenched run-products of sulfide liquids (e.g. O'Neill et al., 1995; Fonseca et al., 2009; Mungall and Brennan, 2014; Bai et al., 2017). Experiment of Helmy et al. (2013b) indicated that the $D_{Pt}^{MSS/Sul}$ decreased from 0.048 to 0.003 while the As increased from 0 ppm to 40 ppm. Non-crystalline Pt-As-rich globules were also recognized in the quenched sulfide liquid (Helmy et al., 2013b). Helmy and Bragagni (2017) found that Ir and Rh formed sulfarsenides whereas Ru formed sulfides at temperature above 1200 °C. These

phenomena demonstrated the existence of Ir-AsS, Rh-AsS and Pt-As pre-nucleation clusters in the sulfide liquid (Helmy et al., 2013b; Helmy and Bragagni, 2017). Thus, in sulfide liquids, (Ir-Rh-Pt)AsS minerals can crystallize through the following reactions (using IrAsS as an example):



Oxygen has a higher electronegativity ($X = 3.44$) than sulfur ($X = 2.58$) so that the valence states of the semimetals in natural sulfide liquids are controlled predominantly by the $f\text{O}_2$. Reducing $f\text{O}_2$ increases the $\text{As}^{1-}/\text{As}^{1+}$ ratio and shifts the equilibrium (3) and (4) to the right, promoting the crystallization of irarsite and decreasing the proportion of Ir occurring in BMS as solid solution. Meanwhile, the activity of As^{1-} increases as elevation of As concentration in the sulfide liquid, resulting in the equilibriums (3) and (4) shifting toward right. Thus, assimilation of As from country rocks may also played an important role in controlling the PGE distribution and will have resulted in the crystallization of IPGE-sulfarsenides in the Creighton deposit, Sudbury (Dare et al., 2010b).

Thus, the sulfide liquid with relatively low $f\text{O}_2$ at Jinchuan had a higher $\text{As}^{1-}/\text{As}^{1+}$ ratio than that at Noril'sk. Although the PGE and semimetal concentrations in natural sulfide liquids (only 0.1 to 1 ppm (10^{-6})) are below the saturation concentrations of PGM (Helmy et al., 2010, 2013a; Bai et al., 2017; Helmy and Bragagni, 2017). Ir-AsS, Rh-AsS and Pt-As pre-nucleation clusters formed in the sulfide liquid at Jinchuan which cannot be incorporated in the crystal lattice of MSS. These pre-nucleation clusters were enriched in or near the migrating MSS grain boundaries and finally reaching oversaturation and nucleation of (Ir-Rh-Pt)AsS occurred. The (Ir-Rh-Pt)AsS crystals were overgrown subsequently by MSS and ISS. Such a process resulted in the poor correlations of Ir with Ru and Rh (Fig. 4A-B), the depletion of Ir and Rh in the pyrrhotite and pentlandite (Fig. 5A) in the Jinchuan massive ores and constant As concentrations in Cu-rich liquids (Fig. S2A; Text S1). The crystallization of euhedral (Ir-Rh-Pt)AsS and PtAs_2 crystals in deposits at Yangliuping and Sudbury were also attributed to the existence of PGE-As(S) pre-nucleation clusters in sulfide liquids (Dare et al., 2010b; Liang et al., 2019). On the other hand, rare Ir-AsS, Rh-AsS and Pt-As pre-nucleation clusters formed in the sulfide liquid at Noril'sk, facilitating Ir and Rh to enter MSS structure whereas the As^{1-} and Pt gradually concentrated in the Cu-rich liquid (Fig. S2D; Text S1). Meanwhile, the crystallization of PtAs_2 at Noril'sk was delayed and it mainly crystallized together with the crystallization of ISS and loop textured pentlandite. This interpretation is supported by the absence of (Ir-Rh-Pt)AsS and PtAs_2 minerals in the quenched sulfide liquid at the vein of Mt. Rudnaya, Noril'sk 1 deposit (Brovchenko et al., 2020), although the sulfide liquid contains comparable As

concentration (15–95 ppm, Brovchenko et al., 2020) with the sulfide liquid at Jinchuan represented approximately by the disseminated ores (Table S1, 51–102 ppm of As in 100% sulfide).

Because the electronegativity of Sn ($X = 1.96$), Sb ($X = 2.05$), Te ($X = 2.10$) and Bi ($X = 2.02$) is lower than As ($X = 2.18$) and they tend to form pre-nucleation clusters with Pd and Pt (Helmy et al., 2010, 2013a, 2013b; Helmy and Bragagni, 2017), they (except As) form Pt-Pd-semimetal minerals at relatively low temperatures (~ 500 °C to 900 °C and up to 1150 °C) compared to IPGE-sulfarsenides (>1200 °C) (Cabri and Laflamme, 1976; Makovicky, 2002; Helmy et al., 2010). Thus, they mainly crystallized together with ISS and loop textured pentlandite (Fig. 7D-E; Figs. S4J, L-O and Q; Table S5B) whereas minor of them crystallized during the crystallization of MSS and then were overgrown by MSS (Fig. 7F; Fig. S4I; Figs. S4K and P; Tables S5).

5.4. Implications on differentiation of PGE in magmatic sulfide deposits

In magmatic sulfide liquids, Rh and the IPGE are compatible with MSS (Barnes et al., 1997; Mungall et al., 2005; Helmy et al., 2010; Liu and Brennan, 2015). There is consensus that Rh and the IPGE differentiated from Pd and Pt by the fractional crystallization of MSS in magmatic sulfide deposits (Naldrett et al., 1996; Mungall et al., 2005; Chen et al., 2013; Duran et al., 2017). However, only a small amount of Ir and Rh are presented in solid solution within BMS in many deposits, such as the deposits at Jinchuan and Yangliuping in China (Chen et al., 2015; Liang et al., 2019) and Sudbury in Canada (Cabri et al., 1984; Dare et al., 2010b). Meanwhile, (Ir-Rh-Pt)AsS crystals could crystallize in sulfide liquids due to local oversaturation and nucleation of the Ir-AsS, Rh-AsS and Pt-As pre-nucleation clusters in or near the migrating MSS grain boundaries. Thus, the differentiation of PGE in magmatic deposits formed from sulfide liquids with a relatively low $f\text{O}_2$ is not controlled by $D_{\text{PGE}}^{\text{MSS}/\text{Sul}}$ alone, but by the simultaneous fractional crystallization of MSS and the (Ir-Rh-Pt)AsS minerals.

6. CONCLUSION

The sulfide liquid at Jinchuan had a lower $f\text{O}_2$ but higher $f\text{S}_2$ compared to the sulfide liquid at Noril'sk. Ir and Rh mainly occur as (Ir-Rh-Pt)AsS minerals within pyrrhotite, pentlandite, chalcopyrite and CGSS or at BMS grain boundaries in the Jinchuan massive ores whereas they mainly occur in BMS as solid solution in the Noril'sk massive ores. The mineral species and morphology and microstructures of PGM suggest that (Ir-Rh-Pt)AsS minerals in the Jinchuan massive ores probably crystallized from the sulfide liquid, whereas the PtS mineral and Pt-Pd-semimetal minerals in the Noril'sk massive ores mainly crystallized during the crystallization of ISS and that loop textured pentlandite formed through a peritectic reaction.

We conclude that the thermodynamic conditions (especially f_{O_2}) control the valence states of semimetals (especially As) thus influencing the behavior of Ir, Rh and Pt during early-stage crystallization of the natural sulfide liquid. At a relatively low f_{O_2} , sulfide liquids have a high As^{n-}/As^{n+} ratio forming abundant Ir-AsS, Rh-AsS and Pt-As pre-nucleation clusters. **These pre-nucleation clusters are incompatible with MSS and trigger the crystallization of (Ir-Rh-Pt)AsS and PtAs₂ crystals due to local oversaturation of these pre-nucleation clusters in or near the migrating MSS grain boundaries.** However, at a relatively high f_{O_2} , As^{n-}/As^{n+} ratio is low in sulfide liquids forming limited Ir-AsS, Rh-AsS and Pt-As pre-nucleation clusters. Thus, Ir and Rh mainly partition into MSS crystal structure together with Os and Ru and the crystallization of sperrylite is delayed. In deposits formed from sulfide liquids with a relatively low f_{O_2} , such as the deposits at Jinchuan and Yangliuping in China and Sudbury in Canada, the fractional crystallization of MSS containing (Ir-Rh-Pt)AsS minerals inclusions resulted in PGE differentiation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

ACKNOWLEDGEMENT

This work was financially supported by grants from the National Key R&D Program of China (2018YFA0702605), Natural Science Foundation of China (41772067 and 41630316 to Song X-Y), Open Funding Project of State Key Laboratory for Mineral Deposits Research, Nanjing University (2018-LAMD-K08). This work was financially supported by grants from the National Key R&D Program of China (2018YFA0702605), Natural Science Foundation of China (41772067 and 41630316 to Song X-Y), Open Funding Project of State Key Laboratory for Mineral Deposits Research, Nanjing University (2018-LAMD-K08). We greatly appreciate Rui Li, Bin Mo, Shao-Hua Dong, Wen-Qin Zheng and Xiang Li for their assistance in SEM and EPMA analyses, Anja Schreiber in TEM foil preparation, Marcus Oelze in discussing PGE analyses using LA-ICP-MS, Danyushevsky, Leonid V. for help in developing the method of PGE analysis, Louis J. Cabri for constructive suggestions and polishing the manuscript, Chris Ballhaus, James Brenan, Yanan Liu and Valeriya D. Brovchenko for useful suggestions. This manuscript benefited from insightful comments from Rubén Piña and one anonymous reviewer, and careful editorial handling by executive editor Jeffrey G. Catalano and associate editor Adam Simon.

APPENDIX A. SUPPLEMENTARY DATA

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.gca.2022.01.017>.

REFERENCES

Bai L. P., Barnes S. J. and Baker D. R. (2017) Sperrylite saturation in magmatic sulfide melts: Implications for formation of PGE-bearing arsenides and sulfarsenides. *Am. Mineral.* **102**, 966–974.

- Ballhaus C. and Ulmer P. (1995) Platinum-group elements in the Merensky Reef: II. Experimental solubilities of platinum and palladium in $Fe_{1-x}S$ from 950 to 450°C under controlled f_{S_2} and f_{H_2} . *Geochim. Cosmochim. Acta* **59**, 4881–4888.
- Ballhaus C., Bockrath C., Wohlgemuth-Ueberwasser C., Laurenz V. and Berndt J. (2006) Fractionation of the noble metals by physical processes. *Contrib. Mineral. Petrol.* **152**(6), 667–684.
- Barnes S. J., Cox R. A. and Zientek M. L. (2006) Platinum-group element, gold, silver and base metal distribution in compositionally zoned sulfide droplets from the Medvezky Creek Mine, Noril'sk, Russia. *Contrib. Mineral. Petrol.* **152**, 187–200.
- Barnes S. J., Godel B., Guerer D., Brenan J. M., Robertson J. and Paterson D. (2013) Sulfide-Olivine Fe-Ni Exchange and the Origin of Anomalously Ni Rich Magmatic Sulfides. *Econ. Geol.* **108**, 1971–1982.
- Barnes S. J., Makovicky E., Makovicky M., Rose-Hansen J. and Karup-Møller S. (1997) Partition coefficients for Ni, Cu, Pd, Pt, Rh, and Ir between monosulfide solid solution and sulfide liquid and the formation of compositionally zoned Ni-Cu sulfide bodies by fractional crystallization of sulfide liquid. *Can. J. Earth Sci.* **34**, 366–374.
- Barnes S. J., Prichard H. M., Cox R. A., Fisher P. C. and Godel B. (2008) The location of the chalcophile and siderophile elements in platinum-group element ore deposits (a textural, microbeam and whole-rock geochemical study): Implications for the formation of the deposits. *Chem. Geol.* **248**, 295–317.
- Barnes S. J. and Ripley E. M. (2016) Highly Siderophile and Strongly Chalcophile Elements in Magmatic Ore Deposits. *Rev. Mineral. Geochem.* **81**, 725–774.
- Barnes S. J., Taranovic V., Schoneveld L. E., Mansur E. T., Le Vaillant M., Dare S., Staude S., Evans N. J. and Blanks D. (2020) The Occurrence and Origin of Pentlandite-Chalcopyrite-Pyrrhotite Loop Textures in Magmatic Ni-Cu Sulfide Ores. *Econ. Geol.* **115**, 1777–1798.
- Brovchenko V. D., Sluzhenikin S. F., Kovalchuk E. V., Kovrigina S. V., Abramova V. D. and Yudovskaya M. A. (2020) Platinum-group Element Enrichment of Natural Quenched Sulfide Solid Solutions, the Noril'sk 1 Deposit, Russia. *Econ. Geol.* **115**, 1343–1361.
- Cabri L. J. (1973) New data on phase relations in the Cu-Fe-S system. *Econ. Geol.* **68**, 443–454.
- Cabri L. J. (1981) Platinum-Group Elements: mineralogy, geology, recovery. *Montreal, Can. Inst. Min. Metall.* **23**, 267 pages.
- Cabri L. J. (2002) The platinum-group minerals. In *The Geology, Geochemistry, Mineralogy, Mineral Beneficiation of the Platinum-Group Elements* (eds. Cabri, L.J.). *Can. Inst. Min., Metall. Petrol.* **54**, 13–129.
- Cabri L. J., Blank H., Goresy A. E., Laflamme J. H. G., Nobiling R., Sizgoric M. B. and Traxel K. (1984) Quantitative trace-element analyses of sulphides from Sudbury and Stillwater by proton microprobe. *Can. Mineral.* **22**, 521–542.
- Cabri L. J., Kelvin M., Yang Z., Jackson S. E. and Altun O. (2017) Application of LA-ICP-MS trace-element analysis for precious metal deportment: a case study of the Kevitsa mine, Finland. *Eur. J. Mineral.* **29**(4), 635–644.
- Cabri L. J. and Laflamme J. H. G. (1976) The mineralogy of the platinum-group elements from some copper-nickel deposits of the Sudbury area, Ontario. *Econ. Geol.* **71**, 1159–1195.
- Cabri L. J., Sylvester P. J., Tubrett M. N., Peregoedova A. and Laflamme J. H. G. (2003) Comparison of LAM-ICP-MS and Micro-PIXE results for palladium and rhodium in selected samples of Noril'sk and Talnakh sulfides. *Can. Mineral.* **41**(2), 321–329.
- Campbell I. H., Czamanske G. K., Fedorenko V. A., Hill R. I. and Stepanov V. (1992) Synchronism of the Siberian Traps and the Permian-Triassic boundary. *Science* **258**, 1760–1763.

- Canali A., Brenan J. and Sullivan N. (2017) Solubility of platinum-arsenide melt and sperrylite in synthetic basalt at 0.1 MPa and 1200° C with implications for arsenic speciation and platinum sequestration in mafic igneous systems. *Geochim. Cosmochim. Acta* **216**, 153–168.
- Chen L. M., Song X. Y., Danyushevsky L. V., Wang Y. S., Tian Y. L. and Xiao J. F. (2015) A laser ablation ICP-MS study of platinum-group and chalcophile elements in base metal sulfide minerals of the Jinchuan Ni–Cu sulfide deposit, NW China. *Ore Geol. Rev.* **65**, 955–967.
- Chen L. M., Song X. Y., Keays R. R., Tian Y. L., Wang Y. S., Deng Y. F. and Xiao J. F. (2013) Segregation and Fractionation of Magmatic Ni–Cu–PGE Sulfides in the Western Jinchuan Intrusion, Northwestern China: Insights from Platinum-group Element Geochemistry. *Econ. Geol.* **108**, 1793–1811.
- Czamanske G. K., Kunilov V. E., Zientek M. L., Cabri L. J., Calk L. C. and Likhachev A. P. (1992) A proton-microprobe study of magmatic sulfide ores from the Noril'sk-Talnakh district, Siberia. *Can. Mineral.* **30**, 249–287.
- Dare S. A. S., Barnes S. J. and Prichard H. M. (2010a) The distribution of platinum-group elements (PGE) and other chalcophile elements among sulfides from the Creighton Ni–Cu–PGE sulfide deposit, Sudbury, Canada, and the origin of palladium in pentlandite. *Miner. Deposita* **45**, 765–793.
- Dare S. A. S., Barnes S. J., Prichard H. M. and Fisher P. C. (2010b) The Timing and Formation of Platinum-Group Minerals from the Creighton Ni–Cu–Platinum-Group Element Sulfide Deposit, Sudbury, Canada: Early Crystallization of PGE-Rich Sulfarsenides. *Econ. Geol.* **105**, 1071–1096.
- Dare S. A. S., Barnes S. J., Prichard H. M. and Fisher P. C. (2014) Mineralogy and Geochemistry of Cu-Rich Ores from the McCreedy East Ni–Cu–PGE Deposit (Sudbury, Canada): Implications for the Behavior of Platinum Group and Chalcophile Elements at the End of Crystallization of a Sulfide Liquid. *Econ. Geol.* **109**, 343–366.
- Deditius A. P., Utsunomiya S., Renock D., Ewing R. C., Ramana C. V., Becker U. and Kesler S. E. (2008) A proposed new type of arsenian pyrite: Composition, nanostructure and geological significance. *Geochim. Cosmochim. Acta* **72**, 2919–2933.
- Distler V. V., Sinyakova E. F. and Kosyakov V. I. (2016) Behavior of noble metals upon fractional crystallization of copper-rich sulfide melts. *Dokl. Earth Sci.* **469**(2), 811–814.
- Duran C. J., Barnes S. J., Pleše P., Prašek M. K., Zientek M. L. and Pagé P. (2017) Fractional crystallization-induced variations in sulfides from the Noril'sk-Talnakh mining district (polar Siberia, Russia). *Ore Geol. Rev.* **90**, 326–351.
- Duran C. J., Barnes S. J., Mansur E. T., Dare S. A. S., Bédard L. P. and Sluzhenikin S. F. (2020) Magnetite chemistry by laser ablation-inductively coupled plasma-mass spectrometry records sulfide fractional crystallization in massive nickel-copper-platinum-group element ores from the Noril'sk-Talnakh mining district (Siberia, Russia): implications for trace element partitioning into magnetite. *Econ. Geol.* **115**(6), 1245–1266.
- Dyuzhikov O. A., Distler V. V., Strunin B. M., Mkrtchyan A. K., Sherman M. L., Sluzhenikin S. F. and Lurye A. M. (1992) Geology and metallogeny of sulfide deposits Noril'sk region USSR. *Econ Geol Monograph* **242**, p.
- Fleet M. E., Chrysosoulis S. L., Stone W. E. and Weisener C. G. (1993) Partitioning of platinum-group elements and Au in the Fe–Ni–Cu–S system: experiments on the fractional crystallization of sulfide melt. *Contrib. Mineral. Petrol.* **115**(1), 36–44.
- Fonseca R. O. C., Campbell I. H., O'Neill H. S. C. and Allen C. M. (2009) Solubility of Pt in sulphide mattes: Implications for the genesis of PGE-rich horizons in layered intrusions. *Geochim. Cosmochim. Acta* **73**, 5764–5777.
- Genkin A.D., Distler V.V., Gladyshev G.D., Filimonova A.A., Evstigneeva T.L., Kovalenker V.A., Laputina I.P., Smirnov A. V. and Grokhovskaya T.L. (1981) *Sulfide Copper–Nickel Ores of the Noril'sk Deposits* (in Russian). Moscow, Nauka.
- Godel B. and Barnes S.-J. (2008) Image analysis and composition of platinum-group minerals in the JM Reef, Stillwater Complex. *Econ. Geol.* **103**, 637–651.
- Godel B., Barnes S. J., Barnes S. J. and Maier W. D. (2010) Platinum ore in three dimensions: Insights from high-resolution X-ray computed tomography. *Geology* **38**, 1127–1130.
- Gottesmann B. and Wirth R. (1997) Pyrrhotite inclusions in dark pigmented apatite from granitic rocks. *Eur. J. Mineral.* **9**, 491–500.
- Harvey J. and Day J. M. D. (2016) Introduction to highly siderophile and strongly chalcophile elements in high temperature geochemistry and cosmochemistry. *Rev. Mineral. Geochem.* **81**, iii–xiv.
- Helmy H. M., Ballhaus C., Fonseca R. O. C. and Nagel T. (2013a) Fractionation of platinum, palladium, nickel, and copper in sulfide–arsenide systems at magmatic temperature. *Contrib. Mineral. Petrol.* **166**, 1725–1737.
- Helmy H. M., Ballhaus C., Fonseca R. O. C., Wirth R., Nagel T. and Tredoux M. (2013b) Noble metal nanoclusters and nanoparticles precede mineral formation in magmatic sulphide melts. *Nat. Commun.* **4**, 2405.
- Helmy H. M., Ballhaus C., Wohlgemuth-Ueberwasser C., Fonseca R. O. C. and Laurenz V. (2010) Partitioning of Se, As, Sb, Te and Bi between monosulfide solid solution and sulfide melt - Application to magmatic sulfide deposits. *Geochim. Cosmochim. Acta* **74**, 6174–6179.
- Helmy H. M. and Botcharnikov R. (2020) Experimental determination of the phase relations of Pt and Pd antimonides and bismuthinides in the Fe–Ni–Cu sulfide systems between 1100 and 700° C. *Am. Miner.* **105**, 344–352.
- Helmy H. M. and Bragagni A. (2017) Platinum-group elements fractionation by selective complexing, the Os, Ir, Ru, Rh–arsenide-sulfide systems above 1020°C. *Geochim. Cosmochim. Acta* **216**, 169–183.
- Hutchinson D. and McDonald I. (2008) Laser ablation ICP-MS study of platinum-group elements in sulphides from the Platreef at Turfspruit, northern limb of the Bushveld Complex, South Africa. *Miner. Depos.* **43**(6), 695–711.
- Keays R. R. and Crocket J. H. (1970) A study of precious metals in the Sudbury nickel irruptive ores. *Econ. Geol.* **65**(4), 438–450.
- Kozyrev S. M., Komarova M. Z., Emelina L. N., Oleshkevich O. I., Yakovleva O. A., Lyalinov D. and Maximov V. I. (2002) The mineralogy and behaviour of the PGM during processing of the Noril'sk-Talnakh PGE–Cu–Ni ores. In *The geology, geochemistry, mineralogy and mineral benification of platinum-group elements* (eds. Cabri L.J.). *Can. Inst. Min., Metall. Petrol.* **54**, 757–792.
- Krivolutskaya N. A. and Sobolev A. I. (2016) *Siberian Traps and Pt–Cu–Ni deposits in the Noril'sk area*. Springer, Switzerland.
- Krivolutskaya N., Tolstykh N., Kedrovskaya T., Naumov K., Kubrakova I., OksanaTutunnik O., Gongalsky B., Kovalchuk E., Magazina L., Ya B. and Yakushev A. (2018) World-class PGE–Cu–Ni Talnakh Deposit: New Data on the Structure and Unique Mineralization of the South-Western Branch. *Minerals* **8**(4), 03–22.
- Kullerud G., Yund R. A., Moh G. H. and Wilson H. D. B. (1969) Phase relations in the Cu–Fe–S, Cu–Ni–S, and Fe–Ni–S systems. Society of Economic Geologists. In *Magmatic ore deposits* (ed. Wilson), pp. 323–343.
- Li C., Ripley E. M. and Naldrett A. J. (2009) A new genetic model for the giant Ni–Cu–PGE sulfide deposits associated with the siberian flood basalts. *Econ. Geol.* **104**, 291–301.

- Li X., Li S., Song B. and Liu D. (2004) SHRIMP U-Pb zircon age of the Jinchuan ultramafic intrusion and its geological significance. *Chin. Sci. Bull.* **49**, 420–422.
- Liang Q. L., Song X. Y., Wirth R., Chen L. M. and Dai Z. H. (2019) Implications of nano- and micrometer-size platinum-group element minerals in base metal sulfides of the Yangliuping Ni-Cu-PGE sulfide deposit, SW China. *Chem. Geol.* **517**, 7–21.
- Lightfoot P. C. and Keays R. R. (2005) Siderophile and chalcophile metal variations in flood basalts from the Siberian trap, Noril'sk region: Implications for the origin of the Ni-Cu-PGE sulfide ores. *Econ. Geol.* **100**(3), 439–462.
- Liu Y. N. and Brennan J. (2015) Partitioning of platinum-group elements (PGE) and chalcogens (Se, Te, As, Sb, Bi) between monosulfide-solid solution (MSS), intermediate solid solution (ISS) and sulfide liquid at controlled fO(2)-fS(2) conditions. *Geochim. Cosmochim. Acta* **159**, 139–161.
- Maciag B. J. and Brennan J. M. (2020) Speciation of arsenic and antimony in basaltic magmas. *Geochim. Cosmochim. Acta* **276**, 198–218.
- Makovicky M., Makovicky E. and Rose-Hansen J. (1986) Experimental studies on the solubility and distribution of platinum-group elements in base-metal sulphides in platinum deposits. In *Metallogeny of Basic and Ultrabasic Rocks* (eds. M. J. Gallagher, R. A. Ixer, C. R. Neary and H. M. Prichard). Institution of Mining and Metallurgy, London, pp. 415–425.
- Makovicky M. (2002) Ternary and Quaternary Phase Systems with PGE. In *The Geology, Geochemistry, Mineralogy, Mineral Beneficiation of the Platinum-Group Elements* (eds. Cabri, L. J.). *Can. Inst. Min., Metall. Petrol.* **54**, 131–175.
- Mansur E. T. and Barnes S. J. (2020) The role of Te, As, Bi, Sn and Sb during the formation of platinum-group-element reef deposits: Examples from the Bushveld and Stillwater Complexes. *Geochim. Cosmochim. Acta* **272**, 235–258.
- Mansur E. T., Barnes S. J. and Duran C. J. (2019a) Textural and compositional evidence for the formation of pentlandite via peritectic reaction: Implications for the distribution of highly siderophile elements. *Geology* **47**(4), 351–354.
- Mansur E. T., Barnes S. J., Duran C. J. and Sluzhenikin S. F. (2019b) Distribution of chalcophile and platinum-group elements among pyrrhotite, pentlandite, chalcopyrite and cubanite from the Noril'sk-Talnakh ores: implications for the formation of platinum-group minerals. *Miner. Deposita* **55**, 1215–1232.
- Mungall J. E., Andrews D. R. A., Cabri L. J., Sylvester P. J. and Tubrett M. (2005) Partitioning of Cu, Ni, Au, and platinum-group elements between monosulfide solid solution and sulfide melt under controlled oxygen and sulfur fugacities. *Geochim. Cosmochim. Acta* **69**, 4349–4360.
- Mungall J. E. and Brennan J. M. (2014) Partitioning of platinum-group elements and Au between sulfide liquid and basalt and the origins of mantle-crust fractionation of the chalcophile elements. *Geochim. Cosmochim. Acta* **125**, 265–289.
- Naldrett A. J. (2004) *Magmatic sulfide deposits: geology, geochemistry, and exploration*. Springer, Berlin.
- Naldrett A. J., Lightfoot P. C., Fedorenko V. A., Doherty W. and Gorbachev N. S. (1992) Geology and geochemistry of intrusions and flood basalts of the Noril'sk region, USSR, with implications for the origin of the Ni-Cu ores. *Econ. Geol.* **335** (4), 975–1004.
- Naldrett A. J., Fedorenko V. A., Asif M., Lin S., Kunilov V. E., Stekhin A. I., Lightfoot P. C. and Gorbachev N. S. (1996) Controls on the composition of Ni-Cu sulfide deposits as illustrated by those at Noril'sk, Siberia. *Econ. Geol.* **91**, 751–773.
- O'Neill H. S. C., Dingwell D. B., Borisov A., Spettel B. and Palme H. (1995) Experimental petrochemistry of some highly siderophile elements at high temperatures, and some implications for core formation and the mantle's early history. *Chem. Geol.* **120**, 255–273.
- Palme H. (2008) Platinum-group elements in cosmochemistry. *Elements* **4**(4), 233–238.
- Pina R., Gervilla F., Barnes S. J., Ortega L. and Lunar R. (2013) Platinum-group elements-bearing pyrite from the Aguablanca Ni-Cu sulphide deposit (SW Spain): a LA-ICP-MS study. *Eur. J. Mineral.* **25**, 241–252.
- Pina R., Gervilla F., Barnes S. J., Ortega L. and Lunar R. (2015) Liquid immiscibility between arsenide and sulfide melts: evidence from a LA-ICP-MS study in magmatic deposits at Serranía de Ronda (Spain). *Miner. Deposita* **50**(3), 265–279.
- Pina R., Gervilla F., Helmy H., Fonseca R. O. C. and Ballhaus C. (2020) Partition behavior of platinum-group elements during the segregation of arsenide melts from sulfide magma. *Am. Mineral.* **105**, 1889–1897.
- Prichard H. M., Knight R. D., Fisher P. C., McDonald I., Zhou M. F. and Wang C. Y. (2013) Distribution of platinum-group elements in magmatic and altered ores in the Jinchuan intrusion, China: an example of selenium remobilization by postmagmatic fluids. *Miner. Deposita* **48**, 767–786.
- Putnis A. (1992) *An introduction to mineral sciences*. Cambridge University Press, New York.
- Sinyakova E. F., Kosyakov V. I. and Borisenko A. S. (2017) Effect of the Presence of As, Bi, and Te on the Behavior of Pt Metals during Fractionation Crystallization of Sulfide Magma. *Dokl. Earth Sci.* **477**, 1422–1425.
- Sixth Geological Unit (1984) *Geology of the Baijiaozi Cu-Ni sulfide deposit*. Geological publishing House, Beijing.
- Song X. Y., Keays R. R., Zhou M. F., Qi L., Ihlenfeld C. and Xiao J. F. (2009) Siderophile and chalcophile elemental constraints on the origin of the Jinchuan Ni-Cu-(PGE) sulfide deposit, NW China. *Geochim. Cosmochim. Acta* **73**, 404–424.
- Song X. Y., Danyushevsky L. V., Keays R. R., Chen L. M., Wang Y. S., Tian Y. L. and Xiao J. F. (2012) Structural, lithological, and geochemical constraints on the dynamic magma plumbing system of the Jinchuan Ni-Cu sulfide deposit, NW China. *Miner. Deposita* **47**, 277–297.
- Song X.-Y., Zhou M.-F., Wang C. Y., Qi L. and Zhang C.-J. (2006) Role of crustal contamination in formation of the Jinchuan intrusion and its world-class Ni-Cu-(PGE) sulfide deposit, northwest China. *Int. Geol. Rev.* **48**, 1113–1132.
- Spiridonov E. M., Kulagov E. A., Serova A. A., Kulikova I. M., Korotaeva N. N., Sereda E. V., Tushentsova I. N., Belyakov S. N. and Zhukov N. N. (2015) Genetic Pd, Pt, Au, Ag, and Rh mineralogy in Noril'sk sulfide ores. *Geology of Ore Deposits* **57**, 402–432.
- Stekhin A. I. (1994) Mineralogical and geochemical characteristics of the Cu-Ni ores of the Oktyabr'sky and Talnakh deposits. In *Proceedings of the Sudbury-Noril'sk Symposium* (eds. Nightfoot P. C. and Naldrett A. J.). Geological Survey, Ontario. Special vol. 5, pp. 217–230.
- Tomkins A. G. (2010) Wetting facilitates late-stage segregation of precious metal-enriched sulfosalt melt in magmatic sulfide systems. *Geology* **38**(10), 951–954.
- Toulmin P. and Barton P. B. (1964) A thermodynamic study of pyrite and pyrrhotite. *Geochim. Cosmochim. Acta* **28**, 641–671.
- Walker R. J., Morgan J. W., Horan M. F., Czamanske G. K., Krogstad E. J., Fedorenko V. A. and Kunilov V. E. (1994) Re-Os isotopic evidence for an enriched-mantle source for the Noril'sk-type, ore-bearing intrusions, Siberia. *Geochim. Cosmochim. Acta* **58**, 4179–4197.
- Wirth R. (2004) Focused Ion Beam (FIB): A novel technology for advanced application of micro- and nanoanalysis in geosciences and applied mineralogy. *Eur. J. Mineral.* **16**, 863–876.

- Wirth R., Reid D. and Schreiber A. (2013) Nanometer-sized platinum-group minerals (PGM) in base metal sulfides: New evidence for an orthomagmatic origin of the Merensky Reef PGE ore deposit, Bushveld Complex, South Africa. *Can. Mineral.* **51**, 143–155.
- Yang S. H., Qu W. J., Tian Y. L., Chen J. F., Yang G. and Du A. D. (2008) Origin of the inconsistent apparent Re-Os ages of the Jinchuan Ni-Cu sulfide ore deposit, China: Post-segregation diffusion of Os. *Chem. Geol.* **247**, 401–418.
- Yin S., Wirth R., Ma C. and Xu J. (2019) The role of mineral nanoparticles at a fluid-magnetite interface: Implications for trace-element uptake in hydrothermal systems. *Am. Mineral.* **104**(8), 1180–1188.
- Yin S., Wirth R., He H., Ma C., Pan J., Xing J., Xu J., Fu J. and Zhang X.-N. (2022) Replacement of magnetite by hematite in hydrothermal systems: A refined redox-independent model. *Earth Planet. Sci. Lett.* **577**, 117282.

Associate editor: Adam Simon